

Introduction to EC2 (Elastic Compute Cloud)



EC2



EC2 Instances

- In corporate data centers, applications are deployed to physical servers
- Where do you deploy applications in the cloud?
 - Rent virtual machines
 - **EC2 instances** - Virtual machines in AWS
 - **EC2 service** - Provision EC2 instances or virtual machines

Understanding Important Features of EC2



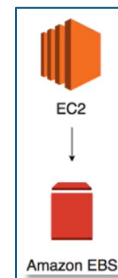
EC2 Instances



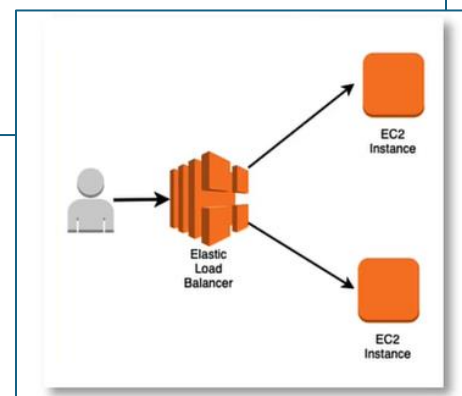
ELB



Amazon EBS



- Create and manage lifecycle of EC2 instances
- **Attach storage** to your EC2 instances
- **Load balancing** for multiple EC2 instances
- **Our Goal:** Play with EC2 instances!



Screenshot as well as personal Notes at bottom

Reviewing Important EC2 Concepts

Feature	Explanation
Amazon Machine Image (AMI)	What operating system and what software do you want on the instance?
Instance Families	Choose the right family of hardware (General purpose or Compute/Storage/Memory optimized or GPU)
Instance Size (t2.nano, t2.micro, t2.small, t2.medium ...)	Choose the right quantity of hardware (2 vCPUs, 4GB of memory)
Elastic Block Store	Attach Disks to EC2 instances (Block Storage)
Security Group	Virtual firewall to control incoming and outgoing traffic to/from AWS resources (EC2 instances, databases etc)
Key pair	Public key and a private key Public key is stored in EC2 instance Private key is stored by customer

Port kya hota hai?

Ek server par multiple services chal sakti hain.

Har service ka ek door number (port) hota hai.

Service	Port
HTTP	80
HTTPS	443
SSH	22
MySQL	3306
PostgreSQL	5432

Firewall technically kya karta hai?

Ye network traffic filter karta hai.

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Ye check karta hai:

Kis IP address se request aa rahi hai?

Kaunse port par aa rahi hai?

Kaunsa protocol use ho raha hai (TCP/UDP)?

Aur fir decide karta hai: ALLOW ☒ ya DENY ☐

AWS mein Security Group kya hota hai?

AWS mein jo EC2 ke saamne firewall lagta hai usko bolte hain: **Security Group**

Ye EC2 level firewall hai.

AWS Security Group mein 2 type ke rule hote, Inbound and Outbound (Ye do cheej sb cloud me hota hai)

Maan lo tum EC2 create karte ho. And Security Group mein rules add karte ho:

Outbound Rules:

Type	Port	Destination
------	------	-------------

All Traffic	All	0.0.0.0/0
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Meaning: Server internet par kisi se bhi baat kar sakta hai.

Inbound Rules:

Type	Port	Source
SSH	22	49.xx.xx.xx/32
HTTP	80	0.0.0.0/0
HTTPS	443	0.0.0.0/0

Iska meaning kya hai?

SSH 22 49.xx.xx.xx/32

Matlab: Sirf tumhare ghar/office ki IP se hi SSH allowed hai.

Baaki duniya: SSH Access = DENIED ☐

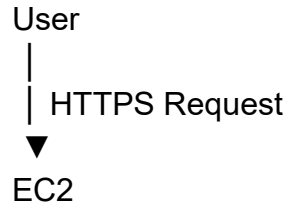
HTTP 80 0.0.0.0/0

Matlab: Internet par koi bhi website access kar sakta hai.

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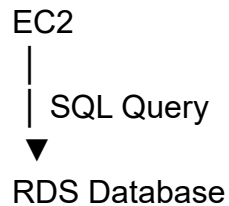
REAL EXAMPLE: Tumne React app deploy kiya EC2 par.

Step 1: User website kholta hai



Ye **Inbound** hai.

Step 2: Server database ko request bhejta hai



Ye **Outbound** hai (EC2 ke perspective se).

Database ka reply



Ye **Inbound** hai (EC2 ke perspective se).

VPC/SUBNETS/IP to client:

Bhai VPC ko **virtual private network** samjho.

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AWS bolta hai: "Yeh Mumbai region ka infrastructure hai. Isme apna network design kar lo."

Tum bolte ho: Mera network: 10.0.0.0/16

AWS: Theek hai. Yeh IP range tumhari. Iska matlab:

10.0.0.0 → 10.0.255.255, ke beech ki IPs tumhare VPC ke andar use hongi.

Toh VPC kahaan laga hota hai?

VPC kisi server pe nahi laga hota.

VPC ek **logical network boundary** hai.

Jaise:

Mumbai Region

├── Company A ka VPC

├── Company B ka VPC

└── Company C ka VPC

VPC ke andar subnets hote hain.

VPC

├── Public Subnet

└── Private Subnet

Subnet VPC ki IP range ko divide karta hai.

Humara: VPC = 10.0.0.0/16

Tumne 2 subnet banaye:

Public Subnet = 10.0.1.0/24

Private Subnet = 10.0.2.0/24

Ye dono VPC ke andar hi hain.

10.0.0.0/16

├── **10.0.1.0/24**

└── **10.0.2.0/24**

Subnet ne IP change nahi kari.

Usne VPC wali badi range ko tukdo mein divide kar diya.

Subnets ke andar resources hote hain.

Screenshot as well as personal Notes at bottom

Public Subnet

- └─ Load Balancer
- └─ Bastion EC2

Private Subnet

- └─ Backend EC2
- └─ Database

Public EC2 me 2 IP kya hoti hai?

Ab maan lo tum EC2 launch karte ho Public Subnet mein.

AWS usko **subnet se private IP** deta hai.

Example:

Public Subnet = 10.0.1.0/24, ye tha right ?

EC2 Private IP = 10.0.1.15, **as ye public subnet me hai toh issi ke andr ki IP milegi**

EC2 ke 2 IPs hote hain:

Private IP: 10.0.1.15. VPC ke andar communication ke liye.

Public IP: 13.232.xx.xx. Internet se access ke liye.

Ye public IP automatically AWS deta hai

Toh EC2 kuch aisa hota hai:

EC2:

Private IP = 10.0.1.15

Public IP = 13.232.xx.xx

Private subnet wali EC2?

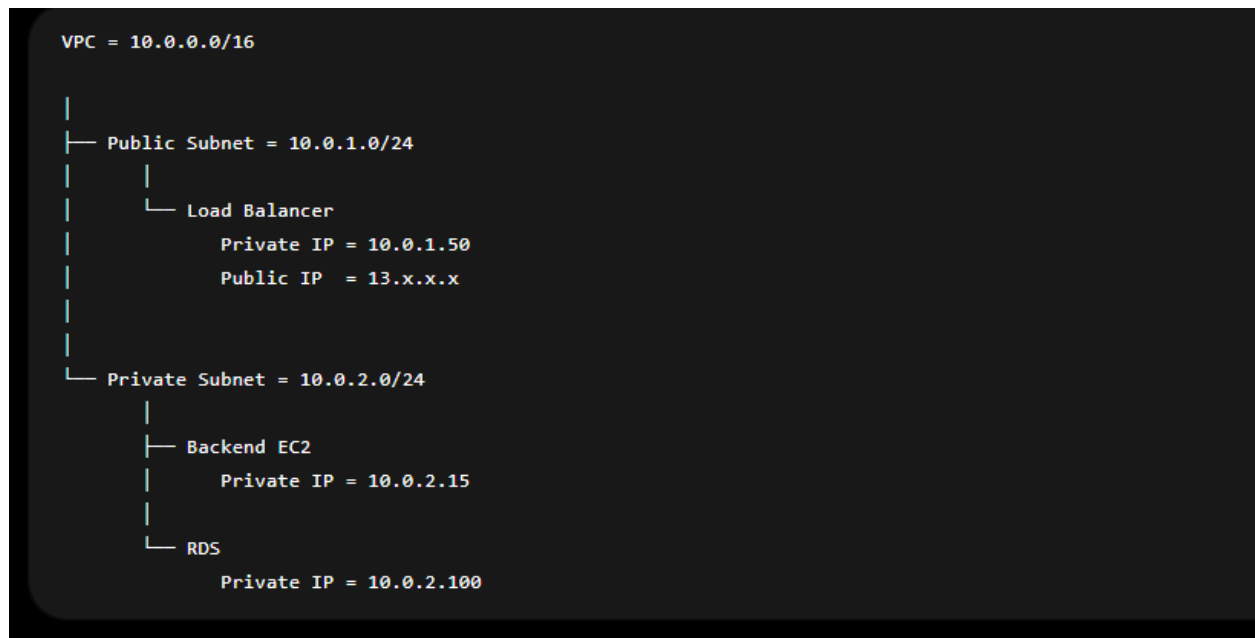
Agar EC2 private subnet mein hai: Then

Private IP = 10.0.2.20

Public IP = NONE

Internet se direct access nahi hoga.

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User request kaise jaati hai?

User: <https://api.example.com>

DNS: api.example.com → 13.x.x.x (load balancer Public IP)

Ye Load Balancer ka public IP hai.

Public EC2 / Load Balancer

AWS alag se **public IP** assign karta hai. Example here => 13.x.x.x

Now Load Balancer se:

13.x.x.x ---- > 10.0.2.15 (Private IP of EC2, as internal communication ho rha)

Backend EC2 ko forward karta hai.

Now Backend se: 10.0.2.15 ---- > 10.0.2.100

Database ko call karta hai.

FIREWALL/SUBNET/SECURITY GROUPS/NACL

Bhai **firewall VPC** pe directly attach nahi hota.

Firewall 2 jagah hota hai:

1. **Subnet pe → NACL**, it will check traffic as soon as req reach at subnet

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2. **Resource pe → Security Group**, it will check traffic as soon as req reach at resources in which security groups attached.

Aur 99% AWS use cases mein jo log "firewall configure karo" bolte hain, wo actually **Security Group configure karne ki baat kar rahe hote hain.**

Toh agar tum poochho:

"Backend EC2 ko sirf Load Balancer access kar sake, internet nahi" —

iska answer hoga:

Backend EC2 ki Security Group mein inbound rule add karo jo sirf Load Balancer Security Group ko allow kare.



Agar NACL bol de: Deny 443 inbound

Then Load Balancer tak request pahunch hi nahi payegi.

Chahe Security Group allow kar raha ho. Becoz NACL firewall traffic checked first at subnet level, then proceed further inside the subnet resources.

Kin resources par Security Group nahi lagta?

✗ S3 Bucket

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S3 apni alag policy system use karta hai:

- Bucket Policies
- IAM Policies
- ACLs

✗ CloudFront

CloudFront Security Groups use nahi karta.